

THE EXACT NON-CENTRAL DISTRIBUTION OF A QUADRATIC FORM IN NORMAL VECTORS

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1. Introduction

Let the columns of X : $p \times n$ be independently normally distributed with covariance matrix Σ and $EX = \mu$, i.e., the density function of X is given by:

$$(1.1) (2\pi)^{-np/2} |\Sigma|^{-n/2} \exp \left\{ -\frac{1}{2} \Sigma^{-1} (X-\mu)(X-\mu)' \right\}.$$

We consider the distribution of the non-central quadratic form $S = XAX'$, where A : $n \times n$ is positive definite.

The distribution of S in the central case, i.e., if $\mu = 0$, has been studied by various authors, including Khatri (1966), Hayakawa (1966) and Shah (1970). Hayakawa (1969) and (1972) obtained the distribution of the characteristic roots of S in the non-central case and if $\Sigma = I$ in terms of polynomials $P_K(T, A)$. We shall make use of a slight generalization of these polynomials of Hayakawa.

The univariate case, i.e., if $p = 1$, has been studied by Ruben (1962) and (1963), Kotz et. al. (1967), Hayakawa (1972) and others. Because of the relative simplicity of the representation of the distribution of S obtained, we also pay attention to the univariate case.

The results on zonal polynomials used may be found in James (1964) and Constantine (1963).

2. The polynomials $P_K(T, A, B)$

Let T : $p \times n$ be a complex matrix, U : $p \times n$ be a real matrix and A : $n \times n$ and B : $p \times p$ be real symmetric matrices. The polynomial $P_K(T, A, B)$ is defined by:

$$\begin{aligned} (2.1) \quad P_K(T, A, B) &= \pi^{-np/2} \exp(\text{tr}(TT')) \int_U \exp \{-UU' - 2iTU'\} C_K(-BUAU') dU \\ &= \pi^{-np/2} \int_U \exp \{-(U+iT)(U+iT)'\} C_K(-BUAU') dU \\ &= E[C_K \{-B(V-iT)A(V-iT)'\}] \end{aligned}$$

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where $C_\kappa(S)$ is the zonal polynomial corresponding to the partition κ of k and the expectation is with respect to the density:

$$(2.2) \pi^{-np/2} \text{etr}(-VV')$$

The integration in (2.1) is over the whole np dimensional Euclidean space.

Lemma 2.1

The j -th cumulant of $\text{tr} \{B(V-iT)A(V-iT)'\}$, i.e., the coefficient of $\theta^j/j!$ in the series expansion of $\ln \phi(\theta) = \ln E[\text{etr} \{ \theta B(V-iT)A(V-iT)' \}]$ is given by:

$$(2.3) c_j = \frac{1}{2} (j-1)! \text{tr}(B^j) \text{tr}(A^j) - j! \text{tr}(B^j T A^j T')$$

where V is distributed according to (2.2).

Proof

Using the same method as e.g. Khatri (1962) or Crowther (1973) it follows that:

$$\begin{aligned} \ln \phi(\theta) &= \frac{1}{2} \sum_{k=1}^{\infty} k^{-1} \theta^k \text{tr}(B^k) \text{tr}(A^k) - \theta \text{tr} B T A T' \\ &\quad - \sum_{k=0}^{\infty} \theta^{k+2} \text{tr}(B^{k+2} T A^{k+2} T'). \end{aligned}$$

This proves the lemma. Using standard results on zonal polynomials, the following properties of the polynomials $P_\kappa(T, A, B)$ are straightforward:

$$(2.4) P_\kappa(O, A, B) = \left(\frac{n}{2}\right)_\kappa (-1)^k C_\kappa(A) C_\kappa(B) / C_\kappa(I)$$

$$(2.5) \sum_{\kappa} P_\kappa(T, A, B) = E[-\text{tr} \{B(V-iT)A(V-iT)'\}]^k$$

the expectation being with respect to the density (2.2). Hence $\sum_{\kappa} P_\kappa(T, A, B)$ may be determined from (2.3), using the usual correspondence between cumulants and moments. If T is real and A and B positive semi-definite, then:

$$(2.6) |P_\kappa(T, A, B)| \leq \left(\frac{n}{2}\right)_\kappa \text{etr}(TT') C_\kappa(A) C_\kappa(B) / C_\kappa(I).$$

In the case where $p = 1$, $T = \underline{t}'$ is a row vector and $B = b$ is a scalar and thus:

$$\begin{aligned} (2.7) P_\kappa(\underline{t}', A, b) &= (-b)^k E[\{(\underline{y}-i\underline{t})' A (\underline{y}-i\underline{t})\}^k] \\ &= (-b)^k p_\kappa(\underline{t}', A), \text{ say.} \end{aligned}$$

In this case the j -th cumulant of $(\underline{y}-i\underline{t})' A (\underline{y}-i\underline{t})$ is given by:

$$(2.8) \quad c_j = \frac{1}{2} (j-1)! \operatorname{tr}(A^j) - j! \operatorname{tr}' A^j \underline{t}.$$

It follows that, for example:

$$(2.9) \quad p_0(\underline{t}', A) = 1$$

$$p_1(\underline{t}', A) = c_1$$

$$p_2(\underline{t}', A) = c_2 + c_1^2$$

$$p_3(\underline{t}', A) = c_3 + 3c_2c_1 + c_1^3$$

$$p_4(\underline{t}', A) = c_4 + 4c_3c_1 + 3c_2^2 + 6c_2c_1^2 + c_1^4$$

etc., the c_j 's being given by (2.8).

3. The distribution of S

Theorem 3.1

Let the density of $X: p \times n$ be given by (1.1). The density function of S is given by:

$$(3.1) \quad \{2^{np/2} \Gamma_p(\frac{n}{2}) |\Sigma|^2 |A|^2\}^{-1} \operatorname{etr}\left(-\frac{1}{2} \Sigma^{-1} \mu \mu'\right) \operatorname{etr}\left(-\frac{q}{2} \Sigma^{-1} S\right) |S|^{(n-p-1)/2} \\ \sum_{k=0}^{\infty} \sum_{\kappa} \left\{ \left(\frac{n}{2}\right)_{\kappa} k! \right\}^{-1} P_{\kappa} \left(-\frac{1}{\sqrt{2}} \Sigma^{-\frac{1}{2}} \mu (I - qA)^{-\frac{1}{2}} A^{-1} - qI, \frac{1}{2} \Sigma^{-\frac{1}{2}} S \Sigma^{-\frac{1}{2}} \right),$$

for $S > 0$ and $q \geq 0$ is an arbitrary constant. If $\mu = 0$, define $(I - qA)^{-\frac{1}{2}} \equiv 0$ and the series is absolutely convergent. If $\mu \neq 0$, $I - qA$ is assumed to be non-singular and the series is at least absolutely convergent if $I - qA > 0$. ($Q^{1/2}$ is supposed to be any symmetric square root of Q . Hence if Q is not positive semi-definite, $Q^{1/2}$ is a complex matrix.)

Proof.

We may assume that $\Sigma = I$. The case where $\Sigma \neq I$ will then follow immediately. The density function of S is:

$$f(S) = (2\pi)^{-np/2} \int_{XAX'=S} \operatorname{etr} \left\{ -\frac{1}{2} (X - \mu)(X - \mu)' \right\} dX \\ = (2\pi)^{-np/2} \int_{XAX'=S} \operatorname{etr} \left\{ -\frac{1}{2} [qXAX' + X(I - qA)X' - 2\mu X' + \mu\mu'] \right\} \\ \cdot \int_U \pi^{-np/2} \operatorname{etr} \left\{ -\left(U - \frac{i}{\sqrt{2}} [X(I - qA)^{\frac{1}{2}} - \mu(I - qA)^{-\frac{1}{2}}] \right) \right\} \\ \left(U - \frac{i}{\sqrt{2}} [X(I - qA)^{\frac{1}{2}} - \mu(I - qA)^{-\frac{1}{2}}] \right)' dU dX$$

since the integral w.r.t. U is unity. By changing the order of integration, it follows that:

$$(3.2) f(S) = (2\pi^2)^{-np/2} \text{etr} \left\{ -\frac{1}{2} \mu \mu' + \frac{1}{2} \mu (I - qA)^{-1} \mu' \right\} \int_U \text{etr} \left\{ -UU' - \sqrt{2} i U(I - qA)^{-\frac{1}{2}} \mu' \right\} \\ \cdot f_{XAX'=S} \text{etr} \left\{ -\frac{1}{2} qXAX' + \sqrt{2} i U(I - qA)^{\frac{1}{2}} X' \right\} dXdU.$$

But

$$\int_{XAX'=S} \text{etr} \left\{ -\frac{1}{2} qXAX' + \sqrt{2} i U(I - qA)^{\frac{1}{2}} X' \right\} dX \\ = \text{etr} \left(-\frac{1}{2} qS \right) \int_{YY'=S} \text{etr} \left\{ \sqrt{2} i U(I - qA)^{\frac{1}{2}} A^{-\frac{1}{2}} Y' \right\} dY \\ = \pi^{np/2} \left\{ \Gamma_p \left(\frac{n}{2} \right) \right\}^{-1} \text{etr} \left(-\frac{1}{2} qS \right) |S|^{(n-p-1)/2} \\ \cdot \sum_{k=0}^{\infty} \sum_{\kappa} \left\{ \left(\frac{n}{2} \right)_{\kappa} k! \right\}^{-1} C_{\kappa} \left(-\frac{1}{2} SU(A^{-1} - qI)U' \right) dU \\ = \pi^{np/2} \left\{ \Gamma_p \left(\frac{n}{2} \right) \right\}^{-1} \text{etr} \left(-\frac{1}{2} qS \right) |S|^{(n-p-1)/2} {}_0F_1 \left(\frac{n}{2}, -\frac{1}{2} SU(A^{-1} - qI)U' \right)$$

where the hypometric series ${}_0F_1$ is defined by Constantine (1963). Substituting this in (3.2) and integrating term by term w.r.t. U , the theorem follows. If $\mu = 0$, the absolute convergence follows using (2.4). If $\mu \neq 0$, the absolute convergence in the case $I - qA > 0$ follows using (2.6).

4. The case $p=1$

Suppose now that $\underline{x}$: $n \times 1$ has a normal distribution with mean $\underline{\mu}$ and covariance matrix I . It follows from theorem (3.1) that the density of $s = \underline{x}'A\underline{x}$, where $A > 0$, is given by:

$$(4.1) \left\{ 2^{n/2} \Gamma \left(\frac{n}{2} \right) |A|^{\frac{1}{2}} \right\}^{-1} e^{-\frac{1}{2} \underline{\mu}' \underline{\mu}} e^{-\frac{1}{2} qs} s^{n/2-1} \sum_{k=0}^{\infty} \left(-\frac{1}{2} s \right)^k \left\{ \left(\frac{n}{2} \right)_k k! \right\}^{-1} \\ \cdot p_k \left(\frac{1}{\sqrt{2}} \underline{\mu}' (I - qA)^{-\frac{1}{2}}, (qA)^{-1} - qI \right).$$

The constant q in (4.1) is subject to similar restrictions as in theorem 3.1. If $q > 0$, the density (4.1) may be given as follows:

$$(4.2) |qA|^{-\frac{1}{2}} e^{-\frac{1}{2} \underline{\mu}' \underline{\mu}} q \sum_{k=0}^{\infty} (-1)^k \left\{ 2^{n/2+k} \Gamma \left(\frac{n+k}{2} \right) k! \right\} e^{-\frac{1}{2} qs} (qs)^{n/2+k-1} \\ \cdot p_k \left(\frac{1}{\sqrt{2}} \underline{\mu}' (I - qA)^{-\frac{1}{2}}, (qA)^{-1} - I \right).$$

Hence the distribution function of s is given by:

$$(4.3) P(s \leq y) = P(qs \leq qy)$$

$$= |qA|^{-\frac{1}{2}} e^{-\frac{1}{2} \underline{\mu}' \underline{\mu}} \sum_{k=0}^{\infty} \frac{(-1)^k}{k!} \Phi_{n+2k}(qy) p_k \left(\frac{1}{\sqrt{2}} \underline{\mu}' (I - qA)^{-\frac{1}{2}} \underline{\mu}, (qA)^{-1} I \right)$$

for $y > 0$ and Φ_{n+2k} is the central χ^2 distribution function with $n+2k$ degrees of freedom. The c_j 's given in (2.8) become in this case.

$$(4.4) c_j = \frac{1}{2} (j-1)! \operatorname{tr} \left\{ \left[(qA)^{-1} I - I \right]^j \right\} \frac{1}{2} j! \underline{\mu}' (I - qA)^{-1} \left\{ (qA)^{-1} I - I \right\}^j \underline{\mu}$$

from which the value of $p_k \left(\frac{1}{\sqrt{2}} \underline{\mu}' (I - qA)^{-\frac{1}{2}} \underline{\mu}, (qA)^{-1} I \right)$ may be found as shown in (2.9).

The constant q may be chosen such that the series converges rapidly.

These representations of the distribution of s are similar to those obtained by Ruben (1962) and Kotz et. al. (1967).

5. The distribution of the characteristic roots of S

Theorem 5.1

The density of the characteristic roots of S is given by:

$$(5.1) \{ 2^{np/2} \Gamma_p \left(\frac{n}{2} \right) \Gamma_p \left(\frac{p}{2} \right) |\Sigma|^{\frac{n}{2}} |A|^{p/2} \}^{-1} \pi^{p^2/2} e^{\operatorname{tr} \left(-\frac{1}{2} \Sigma^{-1} \underline{\mu} \underline{\mu}' \right)} \\ |\Delta|^{(n-p-1)/2} \alpha_p(\Delta) \sum_{k=0}^{\infty} \sum_{\kappa} \left\{ \left(\frac{n}{2} \right)_{\kappa} k! \right\}^{-1} C_{\kappa} \left(\frac{1}{2} \Delta \right) P_{\kappa} \left(\frac{1}{\sqrt{2}} \Sigma^{-\frac{1}{2}} \underline{\mu}, A^{-1}, \Sigma^{-1} \right) / C_{\kappa}(I)$$

where $\Delta = \operatorname{diag}(\lambda_1, \lambda_2, \dots, \lambda_p)$, $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_p > 0$ are the characteristic roots of S and $\alpha_p(\Delta) = \prod_{i < j} (\lambda_i - \lambda_j)$.

Proof.

Let $q = 0$ in theorem 3.1. Let $S = H \Delta H'$ where H is an orthogonal matrix.

Since the jacobian of the transformation is $\alpha_p(\Delta)$ and

$$2^{-p} \int_{O(p)} C_{\kappa} \left(-\frac{1}{2} \Sigma^{-\frac{1}{2}} H \Delta H' \Sigma^{-\frac{1}{2}} U A^{-1} U' \right) dH \\ = \pi^{p^2/2} \{ \Gamma_p \left(\frac{p}{2} \right) \}^{-1} C_{\kappa} \left(\frac{1}{2} \Delta \right) C_{\kappa} \left(-\Sigma^{-1} U A^{-1} U' \right) / C_{\kappa}(I)$$

(see Constantine (1963)), the theorem follows.

The distribution function $F(\lambda_1)$ of the largest characteristic root of S may be obtained from (3.1) using the fact that:

$$\begin{aligned} \int_0^I |S|^{(n-p-1)/2} C_K(-\frac{1}{2} \Sigma^{-\frac{1}{2}} U A^{-1} U' \Sigma^{-\frac{1}{2}} S) ds \\ = \Gamma_p(\frac{n}{2}, \kappa) \Gamma_p(\frac{p+1}{2}) \{ \Gamma_p(\frac{n+p+1}{2}, \kappa) \}^{-1} c^{np/2} C_K(-\frac{1}{2} c \Sigma^{-1} U A^{-1} U') \end{aligned}$$

for any constant $c \geq 0$. Hence:

$$(5.2) \quad F(\lambda_1) = P(S \leq \lambda_1 I)$$

$$\begin{aligned} = \{ 2^{np/2} \Gamma_p(\frac{n+p+1}{2}) |\Sigma|^{n/2} |A|^{p/2} \}^{-1} \Gamma_p(\frac{p+1}{2}) \text{etr}(-\frac{1}{2} \Sigma^{-1} \mu \mu') \\ \cdot \lambda_1^{np/2} \sum_{k=0}^{\infty} \sum_{\kappa} \{ (\frac{n+p+1}{2})_{\kappa} k! \}^{-1} P_{\kappa}(\frac{1}{\sqrt{2}} \Sigma^{-\frac{1}{2}} \mu, A^{-1} \frac{1}{2} \lambda_1 \Sigma^{-1}). \end{aligned}$$

for $\lambda_1 > 0$.

Hayakawa (1969) and (1972) derived (5.1) and (5.2) in the case where $\Sigma = I$.

6. Calculation of the polynomials $P_{\kappa}(T, A, B)$

The polynomials $P_{\kappa}(T, A, B)$ are tedious to calculate. However, we calculate the first few. $P_{\kappa}(T, A, B)$ may be written as follows:

$$(6.1) \quad P_{\kappa}(T, A, B) = \pi^{-\frac{np}{2}} (-1)^k \int_U \text{etr} \{ -(U + iHT)(U + iHT)' \} C_{\kappa}(B_1 U A U') dU$$

where $HBH' = B_1$, H is orthogonal and B_1 is diagonal. By writing $C_{\kappa}(B_1 U A U')$ in terms of products of sums of powers of the characteristic roots of $B_1 U A U'$, the value of $P_{\kappa}(T, A, B)$ may be found by using (2.3). It then follows that:

$$(6.2) \quad P_{(1)}(T, A, B) = -E \text{tr}(B_1 U A U') = -\frac{1}{2} \text{tr} B \text{tr} A + \text{tr}(BTAT')$$

and the expectation is w.r.t. the density:

$$(6.3) \quad \pi^{-np/2} \text{etr} \{ -(U + iHT)(U + iHT)' \}.$$

Also by using (2.3), it follows that:

$$E \text{tr}(B_1 U A U')^2 = E \sum_i \sum_j b_i b_j (u_i' A u_j)^2$$

where $U = \begin{pmatrix} \underline{u}'_1 \\ \vdots \\ \underline{u}'_p \end{pmatrix}$ and $B_1 = \text{diag}(b_1, \dots, b_p)$. Hence

$$E \text{tr}(B_1 U A U')^2 = E \sum_i b_i^2 (\underline{u}'_i A \underline{u}_i)^2 + E \sum_{i \neq j} b_i b_j (\underline{u}'_i A \underline{u}_j)^2$$

Let $\underline{h}'_i$ be the i -th row of H . Then:

$$E(\underline{u}'_i A \underline{u}_j)^2 = \frac{1}{2} \text{tr} A^2 + \frac{1}{4} (\text{tr} A)^2 - \text{tr} A \underline{h}'_i T A T' \underline{h}_i - 2 \underline{h}'_i T A^2 T' \underline{h}_i + (\underline{h}'_i T A T' \underline{h}_i)^2$$

$$E \{(\underline{u}'_i A \underline{u}_j \underline{u}'_j A \underline{u}_i) / \underline{u}_j\} = \frac{1}{2} \underline{u}'_i A^2 \underline{u}_j - \underline{u}'_j A T' \underline{h}_i \underline{h}'_i T A \underline{u}_j$$

and

$$E(\underline{u}'_i A \underline{u}_j)^2 = \frac{1}{4} \text{tr} A^2 - \underline{h}'_j T A^2 T' \underline{h}_j + (\underline{h}'_j T A T' \underline{h}_j)^2.$$

It follows that:

$$(6.4) \quad E \text{tr}(B_1 U A U')^2 = \frac{1}{4} \text{tr} A^2 \text{tr} B^2 + \frac{1}{4} (\text{tr} A)^2 \text{tr} B^2 + \frac{1}{4} \text{tr} A^2 (\text{tr} B)^2 \\ - \text{tr} A \text{tr} B^2 T A T' - \text{tr} B \text{tr} B T A^2 T' - \text{tr} B^2 T A^2 T' + \text{tr}(B T A T')^2$$

and by using (2.3):

$$(6.5) \quad E(\text{tr} B_1 U A U')^2 = \frac{1}{2} \text{tr} A^2 \text{tr} B^2 + \frac{1}{4} (\text{tr} A)^2 (\text{tr} B)^2 - \text{tr} A \text{tr} B \text{tr} B T A T' \\ - 2 \text{tr} B^2 T A^2 T' + (\text{tr} B T A T')^2.$$

Hence:

$$(6.6) \quad P_{(2)}(T, A, B) = \frac{2}{3} E(\text{tr} B_1 U A U')^2 - \frac{2}{3} E \text{tr}(B_1 U A U')^2$$

$$(6.7) \quad P_{(1,1)}(T, A, B) = \frac{1}{3} E(\text{tr} B_1 U A U')^2 + \frac{2}{3} E \text{tr}(B_1 U A U')^2.$$

The polynomials $P_k(T, A, B)$ for larger values of k may be calculated in a similar way.

We list these for $k = 3$:

$$(6.8) \quad E \operatorname{tr}(B_1 UAU')^3$$

$$\begin{aligned}
&= \frac{1}{2} \operatorname{tr} A^3 \operatorname{tr} B^3 + \frac{1}{8} (\operatorname{tr} A)^3 \operatorname{tr} B^3 + \frac{1}{8} \operatorname{tr} A^3 (\operatorname{tr} B)^3 + \frac{3}{8} \operatorname{tr} A^2 \operatorname{tr} A \operatorname{tr} B^3 \\
&+ \frac{3}{8} \operatorname{tr} A^3 \operatorname{tr} B^2 \operatorname{tr} B + \frac{3}{8} \operatorname{tr} A^2 \operatorname{tr} A \operatorname{tr} B^2 \operatorname{tr} B \\
&- \frac{3}{4} \operatorname{tr} A^2 \operatorname{tr} B \operatorname{tr} B^2 \operatorname{tr} TAT' - \frac{3}{4} \operatorname{tr} A \operatorname{tr} B^2 \operatorname{tr} BTA^2 T' \\
&- \frac{3}{4} \operatorname{tr} A^2 \operatorname{tr} B^3 \operatorname{tr} TAT' - \frac{3}{4} \operatorname{tr} B^2 \operatorname{tr} BTA^3 T' \\
&- \frac{3}{4} (\operatorname{tr} A)^2 \operatorname{tr} B^3 \operatorname{tr} TAT' - \frac{3}{4} (\operatorname{tr} B)^2 \operatorname{tr} BTA^3 T' - \frac{3}{4} \operatorname{tr} A \operatorname{tr} B \operatorname{tr} B^2 \operatorname{tr} TA^2 T' \\
&- \frac{3}{2} \operatorname{tr} A \operatorname{tr} B^3 \operatorname{tr} TA^2 T' - \frac{3}{2} \operatorname{tr} B \operatorname{tr} B^2 \operatorname{tr} TA^3 T' \\
&+ \frac{3}{2} \operatorname{tr} A \operatorname{tr} B^2 \operatorname{tr} TAT' \operatorname{tr} BTAT' + \frac{3}{2} \operatorname{tr} B \operatorname{tr} BTA^2 T' \operatorname{tr} BTAT' \\
&+ \frac{3}{2} \operatorname{tr} B^2 \operatorname{tr} TAT' \operatorname{tr} BTA^2 T' - \frac{3}{2} \operatorname{tr} B^2 \operatorname{tr} TAT' \operatorname{tr} BTA^2 T' \\
&- \frac{9}{2} \operatorname{tr} B^2 \operatorname{tr} TA^2 T' \operatorname{tr} BTAT' - 3 \operatorname{tr} B^3 \operatorname{tr} TA^3 T' - \operatorname{tr}(BTAT')^3
\end{aligned}$$

$$(6.9) \quad E \operatorname{tr}(B_1 UAU')^2 \operatorname{tr} B_1 UAU'$$

$$\begin{aligned}
&= \frac{1}{2} \operatorname{tr} A^3 \operatorname{tr} B^3 + \frac{1}{2} \operatorname{tr} A^2 \operatorname{tr} A \operatorname{tr} B^3 + \frac{1}{2} \operatorname{tr} A^3 \operatorname{tr} B^2 \operatorname{tr} B + \frac{1}{8} \operatorname{tr} A^2 \operatorname{tr} A (\operatorname{tr} B)^3 \\
&+ \frac{1}{8} (\operatorname{tr} A)^3 \operatorname{tr} B^2 \operatorname{tr} B + \frac{1}{8} \operatorname{tr} A^2 \operatorname{tr} A \operatorname{tr} B^2 \operatorname{tr} B \\
&- \frac{1}{4} \operatorname{tr} A^2 \operatorname{tr} B^2 \operatorname{tr} BTAT' - \frac{1}{4} \operatorname{tr} A^2 (\operatorname{tr} B)^2 \operatorname{tr} BTAT' - \frac{1}{4} (\operatorname{tr} A)^2 \operatorname{tr} B^2 \operatorname{tr} BTAT' \\
&+ \frac{1}{4} \operatorname{tr} A^2 \operatorname{tr} B \operatorname{tr} B^2 \operatorname{tr} TAT' + \frac{1}{4} \operatorname{tr} A \operatorname{tr} B^2 \operatorname{tr} BTA^2 T' \\
&- \frac{1}{2} \operatorname{tr} A (\operatorname{tr} B)^2 \operatorname{tr} BTA^2 T' - \frac{1}{2} (\operatorname{tr} A)^2 \operatorname{tr} B \operatorname{tr} B^2 \operatorname{tr} TAT' \\
&+ \frac{1}{2} \operatorname{tr} A \operatorname{tr} B \operatorname{tr}(BTAT')^2 - \frac{1}{2} \operatorname{tr} A \operatorname{tr} B \operatorname{tr} B^2 \operatorname{tr} TA^2 T' \\
&- \operatorname{tr} A^2 \operatorname{tr} B^3 \operatorname{tr} TAT' - \operatorname{tr} B^2 \operatorname{tr} BTA^3 T' \\
&+ \operatorname{tr} A \operatorname{tr} B^2 \operatorname{tr} TAT' \operatorname{tr} BTAT' + \operatorname{tr} B \operatorname{tr} BTA^2 T' \operatorname{tr} BTAT'
\end{aligned}$$

$$\begin{aligned}
& -2 \operatorname{tr} A \operatorname{tr} B^3 T A^2 T' - \operatorname{tr} B \operatorname{tr} B^2 T A^3 T' \\
& + \operatorname{tr} B^2 T A^2 T' \operatorname{tr} B T A T' - \operatorname{tr}(B T A T')^2 \operatorname{tr} B T A T' \\
& + 4 \operatorname{tr} B^2 T A^2 T' B T A T' - 3 \operatorname{tr} B^3 T A^3 T'
\end{aligned}$$

$$(6.10) \quad E(\operatorname{tr} B_1 U A U')^3$$

$$\begin{aligned}
& = \operatorname{tr} A^3 \operatorname{tr} B^3 + \frac{1}{8} (\operatorname{tr} A)^3 (\operatorname{tr} B)^3 + \frac{3}{4} \operatorname{tr} A^2 \operatorname{tr} A \operatorname{tr} B^2 \operatorname{tr} B \\
& - \frac{3}{2} \operatorname{tr} A^2 \operatorname{tr} B^2 \operatorname{tr} B T A T' - \frac{3}{4} (\operatorname{tr} A)^2 (\operatorname{tr} B)^2 \operatorname{tr} B T A T' \\
& - 3 \operatorname{tr} A \operatorname{tr} B \operatorname{tr} B^2 T A^2 T' + \frac{3}{2} \operatorname{tr} A \operatorname{tr} B (\operatorname{tr} B T A T')^2 \\
& + 6 \operatorname{tr} B^2 T A^2 T' \operatorname{tr} B T A T' - 6 \operatorname{tr} B^3 T A^3 T' - (\operatorname{tr} B T A T')^3
\end{aligned}$$

$$\begin{aligned}
(6.11) \quad P_{(3)}(T, A, B) & = -\frac{1}{15} E(\operatorname{tr} B_1 U A U')^3 - \frac{2}{5} E \operatorname{tr}(B_1 U A U')^2 \operatorname{tr} B_1 U A U' \\
& - \frac{8}{15} E \operatorname{tr}(B_1 U A U')^3
\end{aligned}$$

$$\begin{aligned}
(6.12) \quad P_{(2,1)}(T, A, B) & = -\frac{3}{5} E(\operatorname{tr} B_1 U A U')^3 - \frac{3}{5} E \operatorname{tr}(B_1 U A U')^2 \operatorname{tr} B_1 U A U' \\
& + \frac{6}{5} E \operatorname{tr}(B_1 U A U')^3
\end{aligned}$$

$$(6.13) \quad P_{(1,1,1)}(T, A, B)$$

$$= -\frac{1}{3} E(\operatorname{tr} B_1 U A U')^3 + E \operatorname{tr}(B_1 U A U')^2 \operatorname{tr} B_1 U A U' - \frac{2}{3} E \operatorname{tr}(B_1 U A U')^3$$

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